

# Inflation

Extreme expansion in the very earliest universe

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# Looking back in time: 100,000 years

The Milky Way



# Looking back in time: 300 Million years

The Coma Cluster



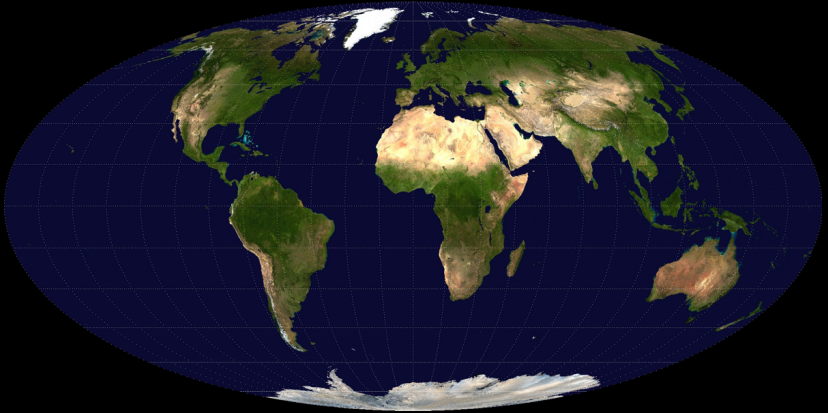
# Looking back in time: 13.7 Billion years

Hubble Extreme Deep Field



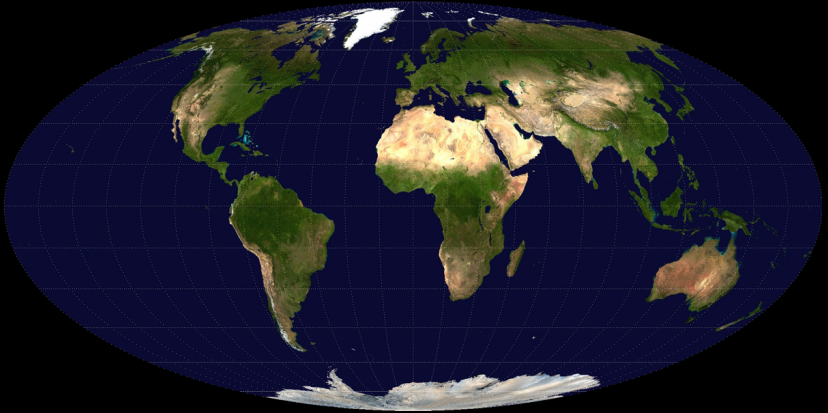


# The Cosmic Microwave Background



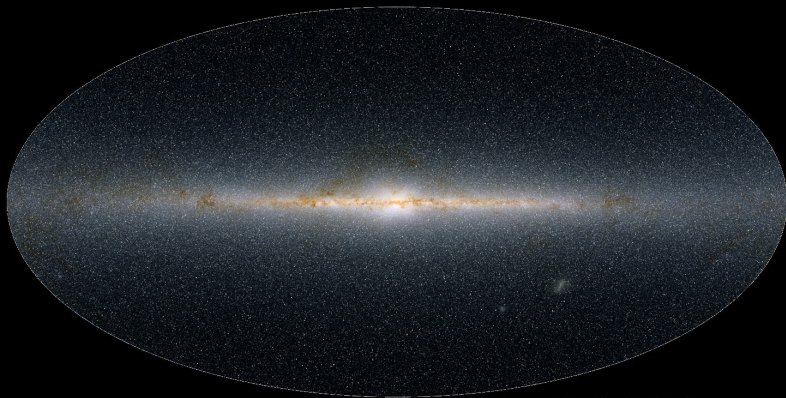
- ▶ A map of the Earth (Mollweide projection).

# The Cosmic Microwave Background



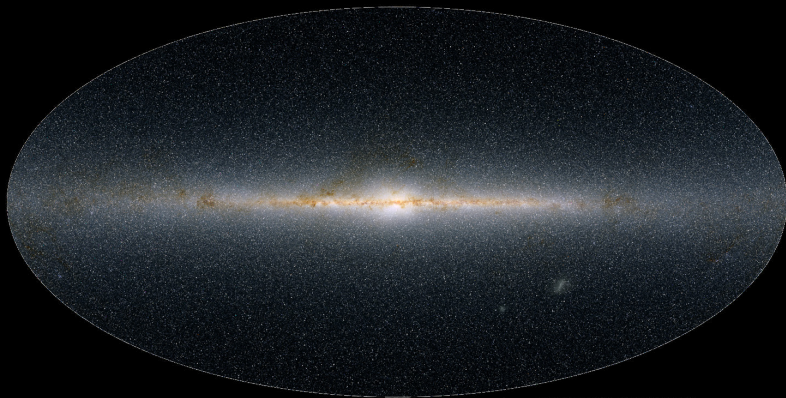
- ▶ A map of the Earth (Mollweide projection).
- ▶ Consider the same map, but looking out at the night sky.

# The Cosmic Microwave Background



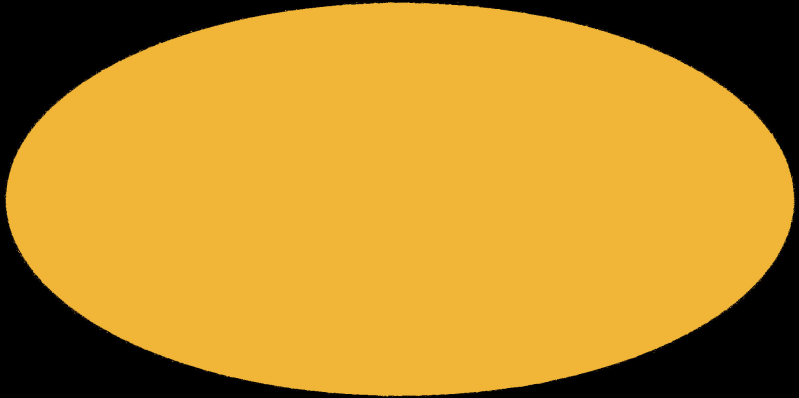
- ▶ This image is taken in the infra-red waveband.

# The Cosmic Microwave Background



- ▶ This image is taken in the infra-red waveband.
- ▶ Now look in the microwave band.

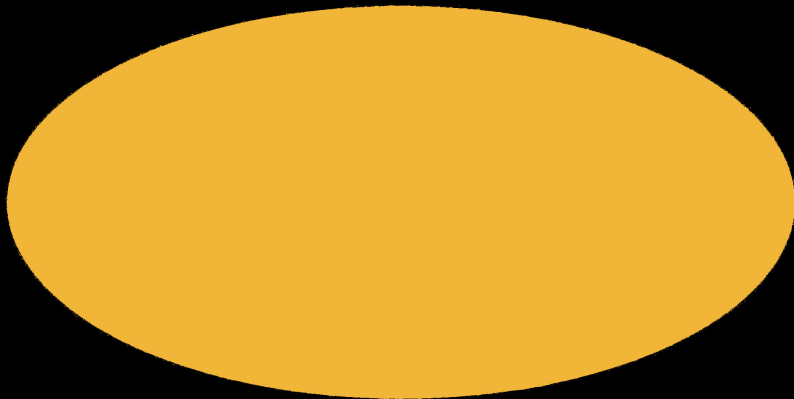
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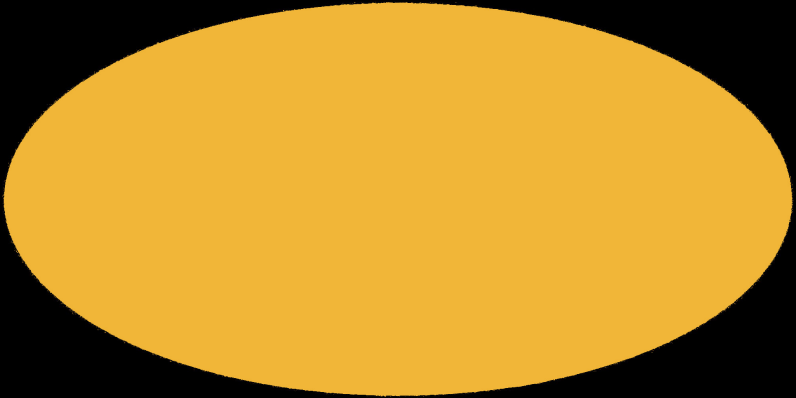


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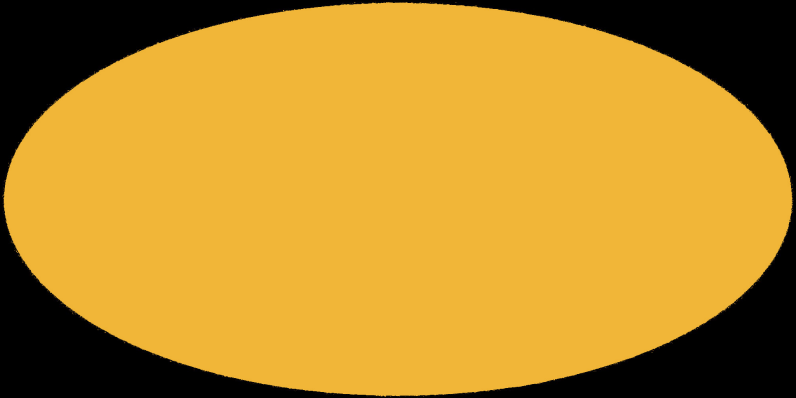
- ▶ The sky is filled with the Cosmic Microwave Background radiation (CMB).

# The Cosmic Microwave Background



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- ▶ The CMB has a temperature of 2.7260 K.

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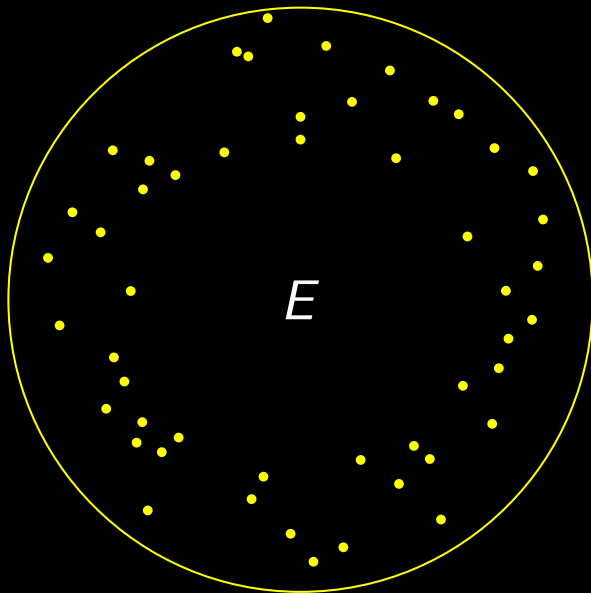
- ▶ The sky is filled with the Cosmic Microwave Background radiation (CMB).
- ▶ The CMB has a temperature of 2.7260 K.
- ▶ It is *too* uniform.

# The Cosmic Microwave Background

The first light of the universe

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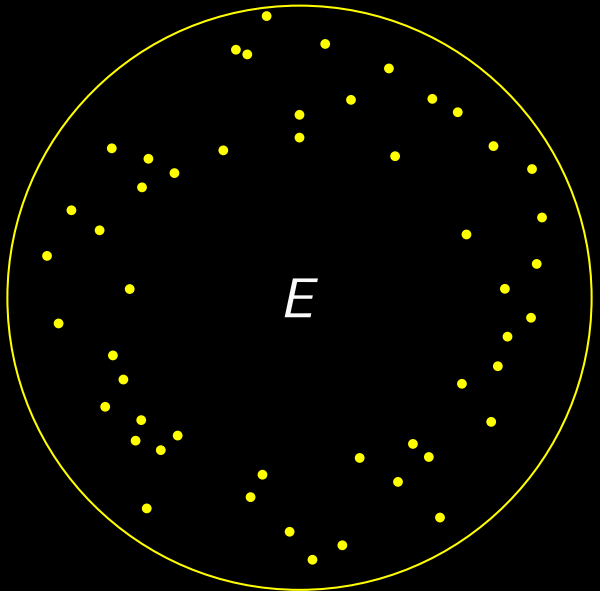
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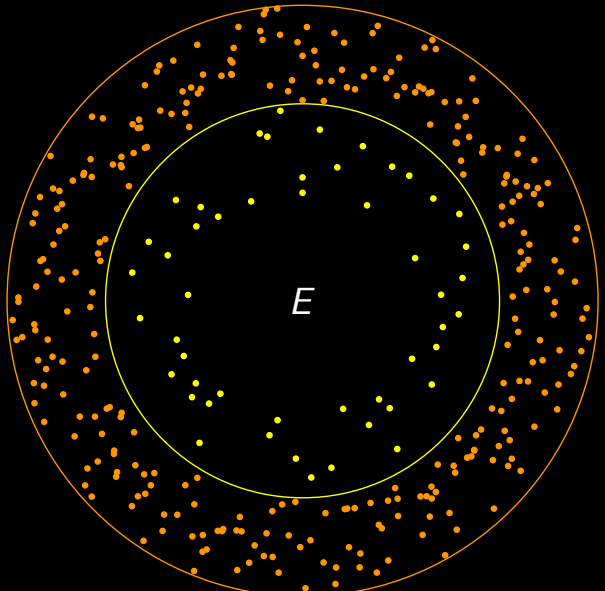
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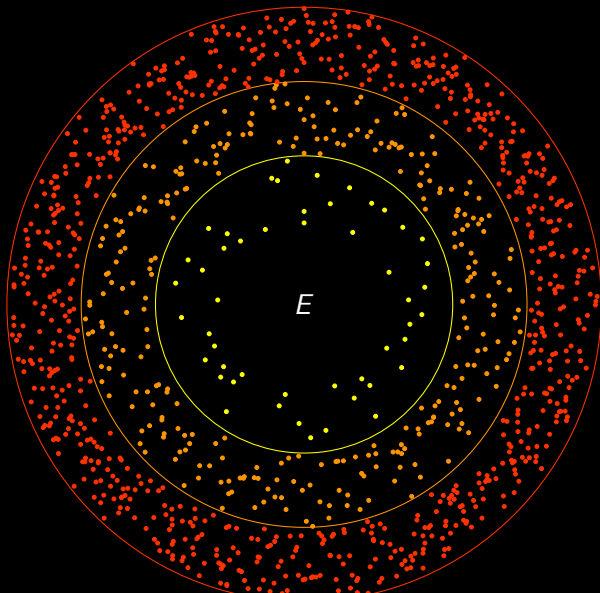




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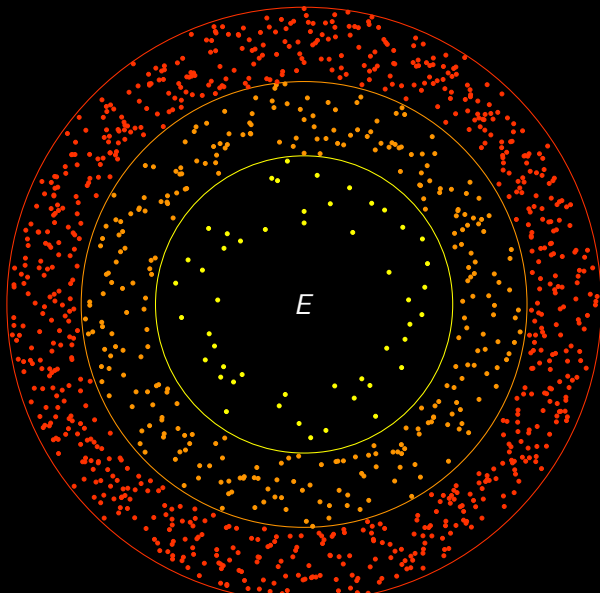
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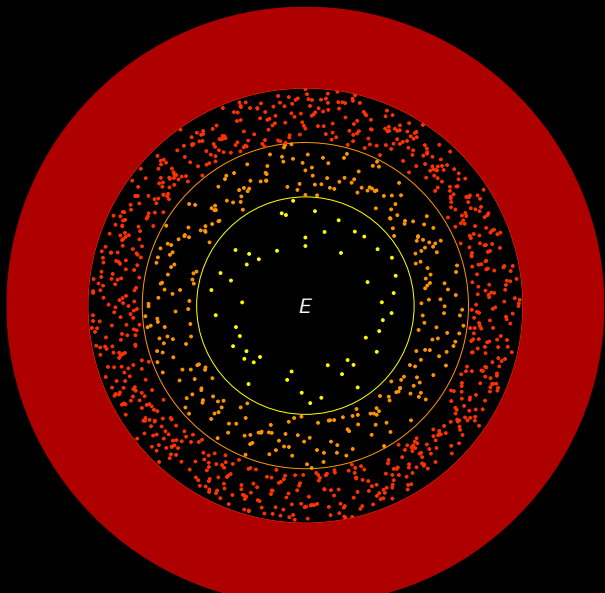
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- ▶ Eventually, universe becomes 'solid'



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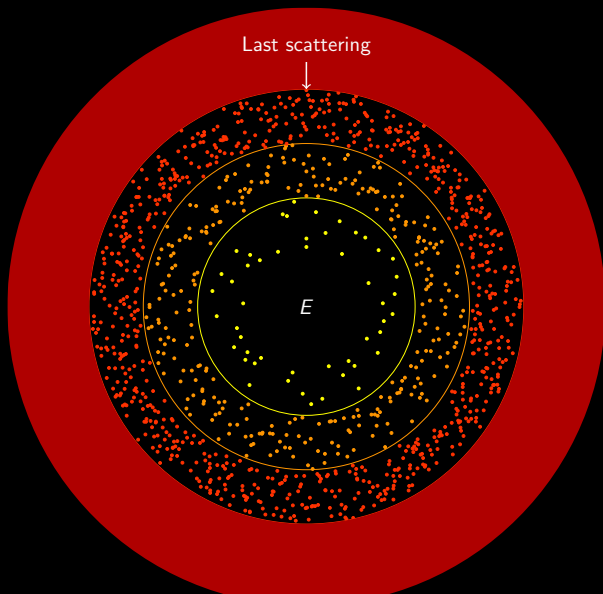
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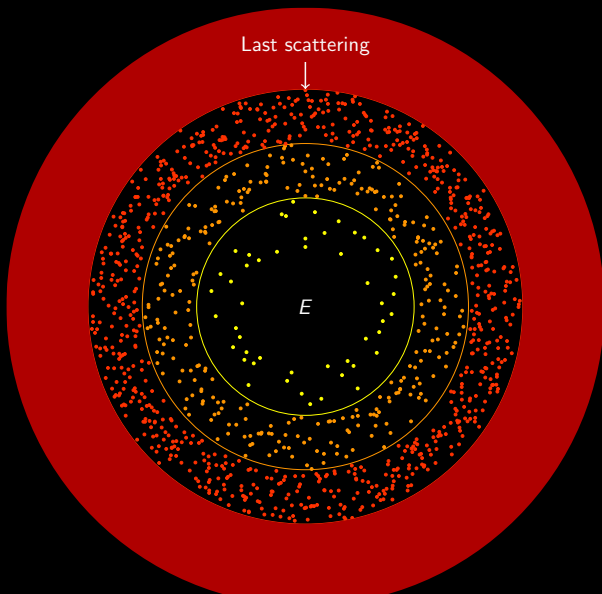
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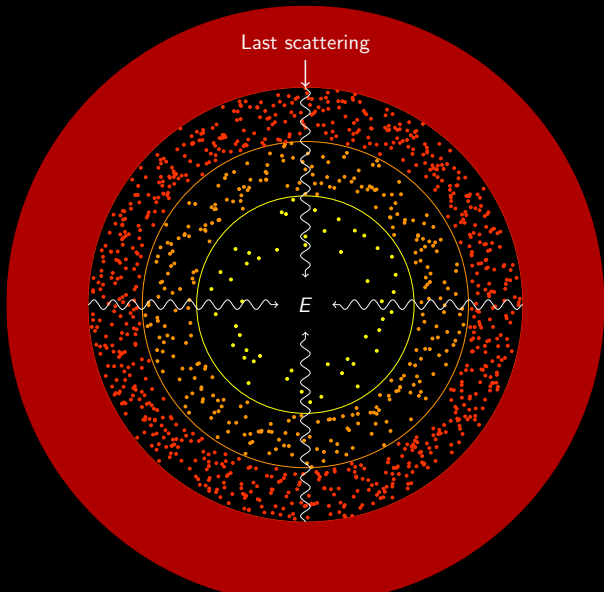
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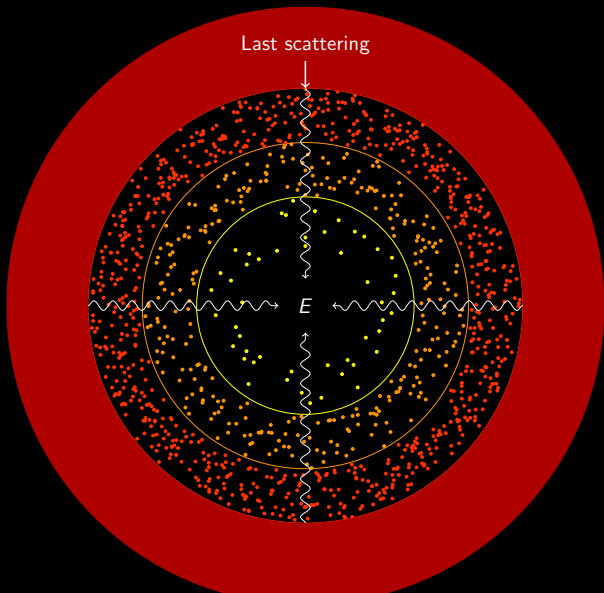
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# The Cosmic Microwave Background

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- This is observed as the CMB



# The issue with uniformity

## Thermal equilibrium



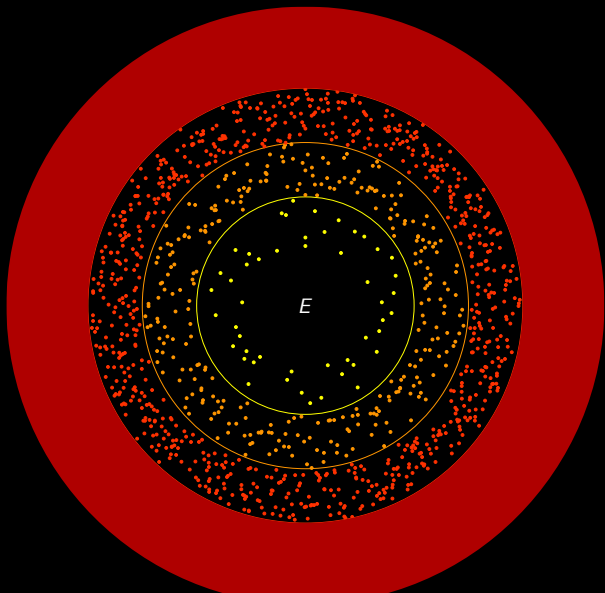
- ▶ The early universe appears to be in *thermal equilibrium*.



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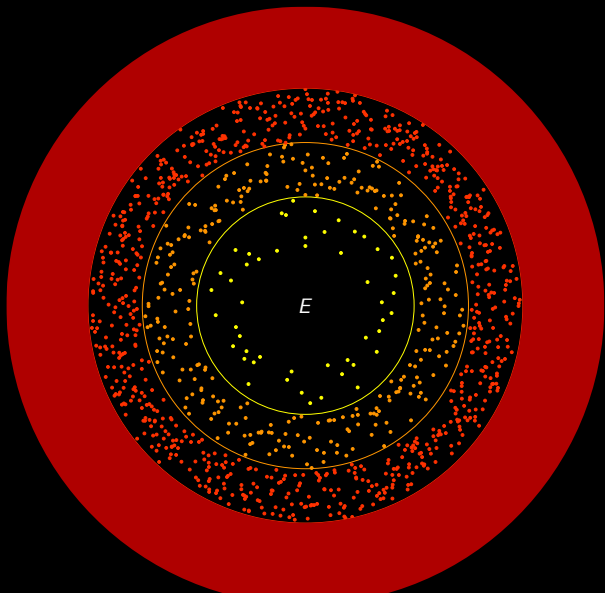
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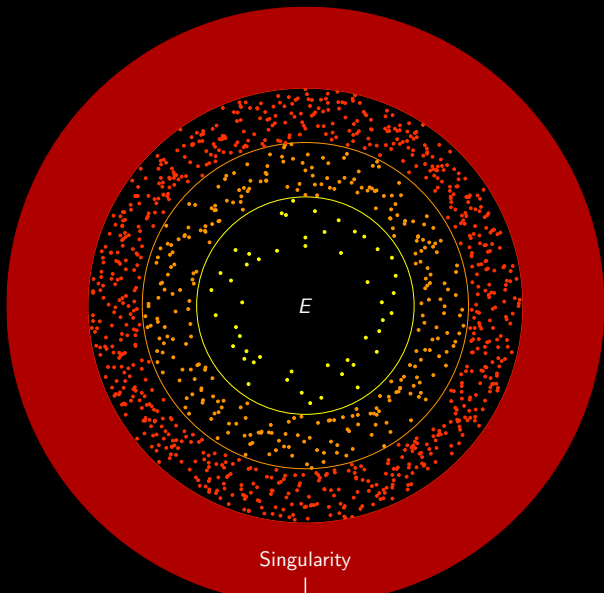
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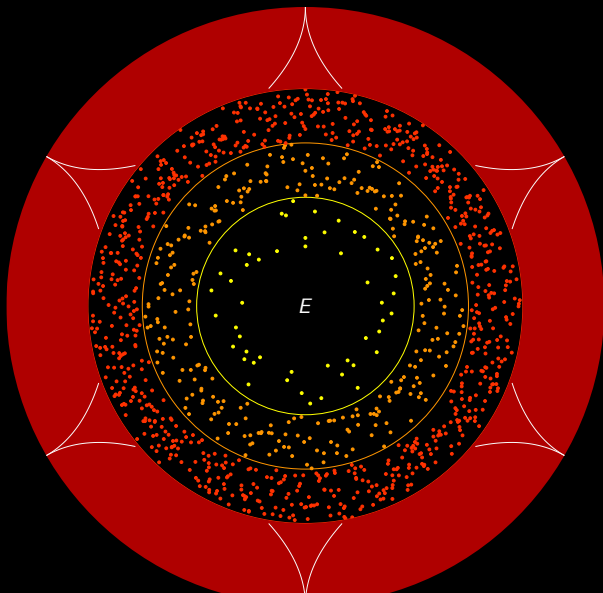
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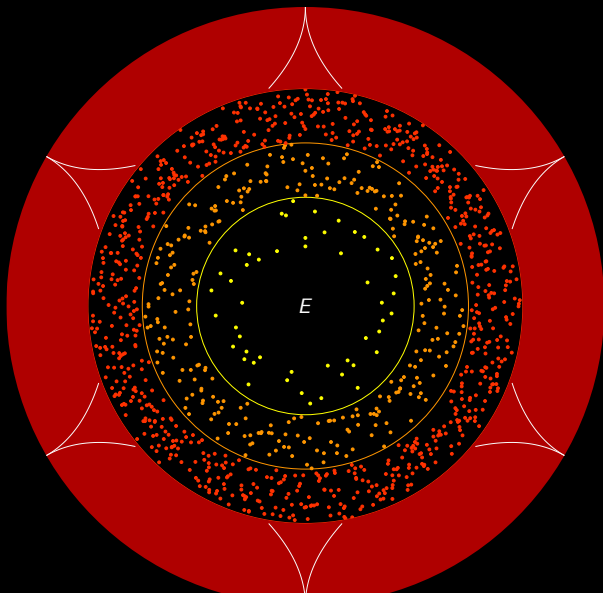
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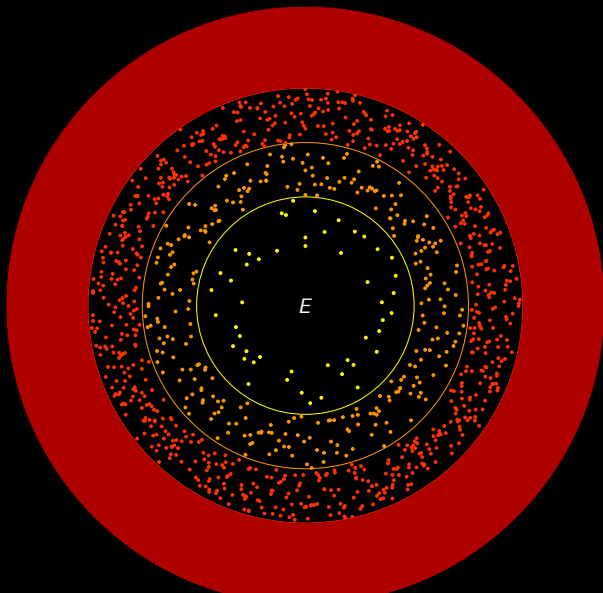
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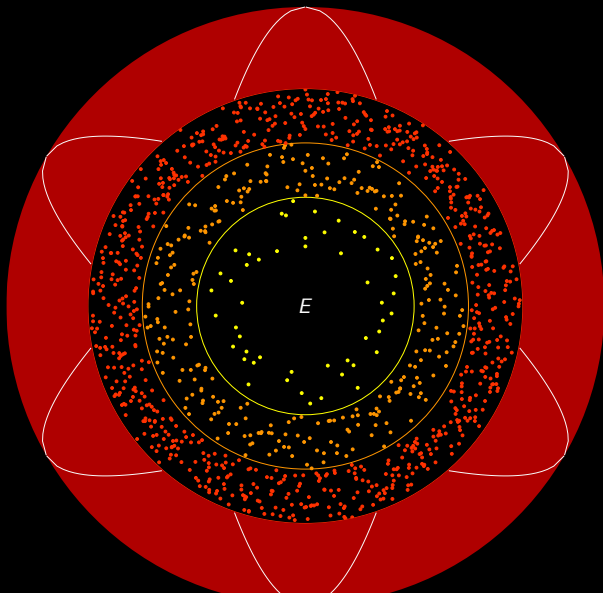
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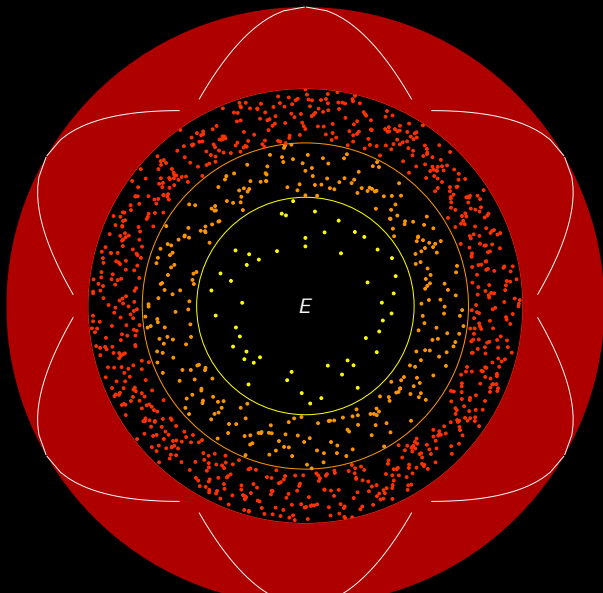
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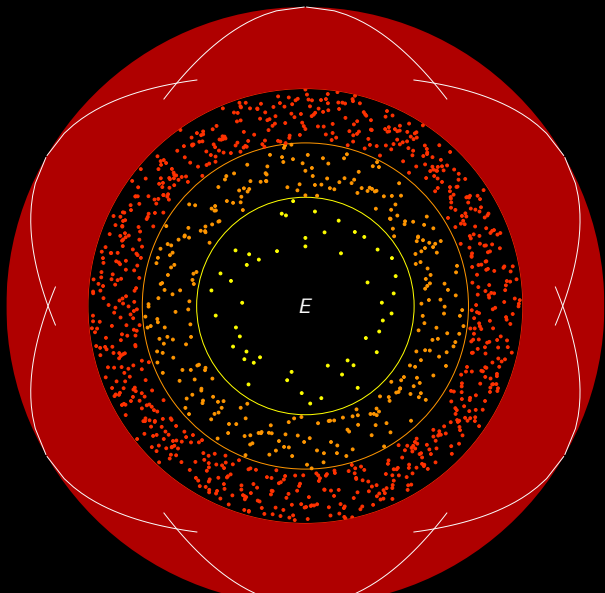




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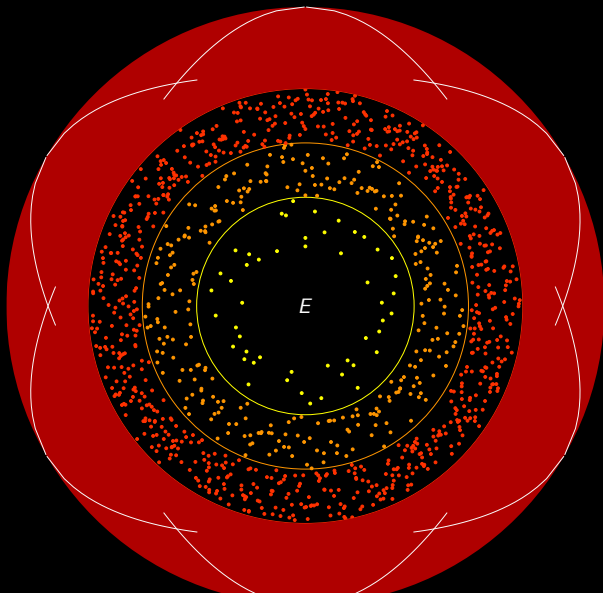
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# The Cosmic Microwave Background

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- ▶ How causally connected is the CMB?
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- ▶ Early acceleration increases patch size.
- ▶ Inflation!



# Inflation

A convenient explanation

1. The CMB is far too uniform

# Inflation

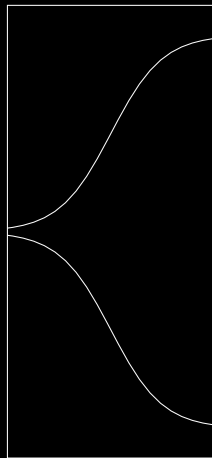
A convenient explanation

1. The CMB is far too uniform
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# Inflation

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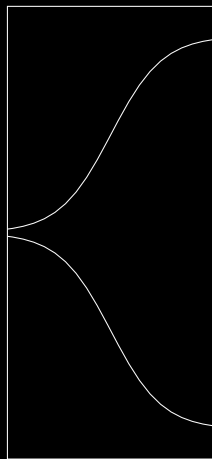
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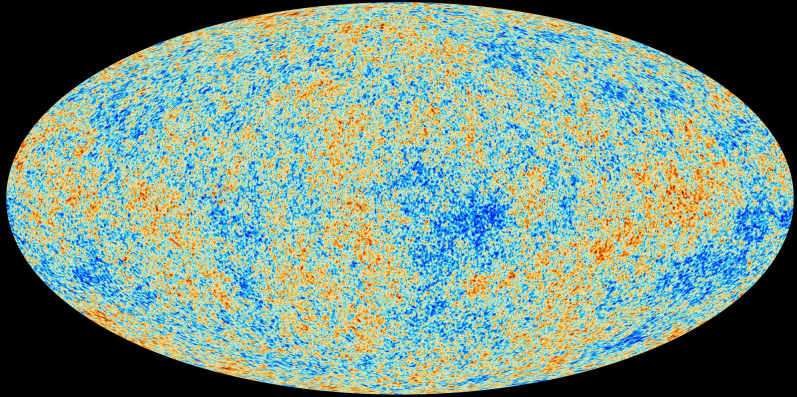
# Inflation

## A convenient explanation

1. The CMB is far too uniform
2. It looks like it expanded incredibly rapidly “long before” the surface of last scattering.
3. We call this accelerated expansion “Inflation”

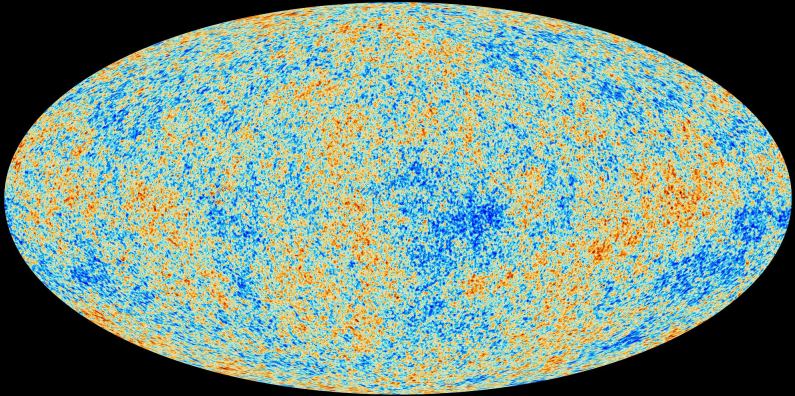


# Anisotropies in the CMB



- ▶ The CMB has a temperature of  $2.7260 \pm 0.0013$  K:
- ▶ The *Theory of Inflation* predicts these inhomogeneities,

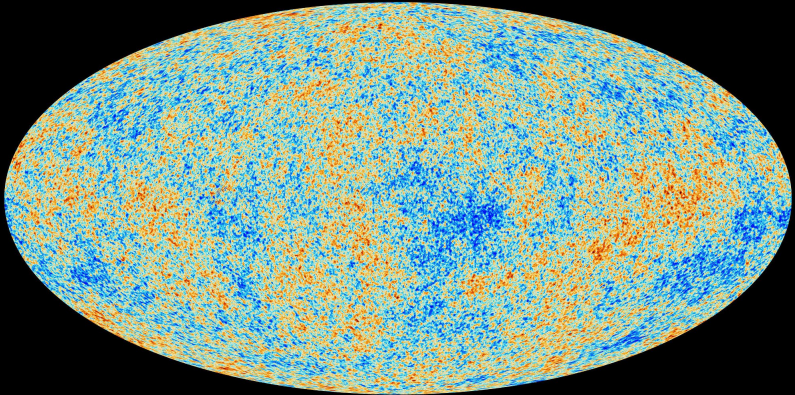
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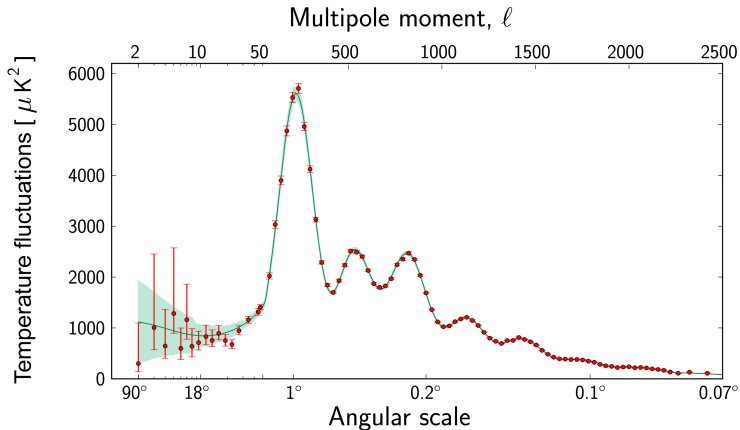
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- ▶ These are the seeds of structure formation in the universe.

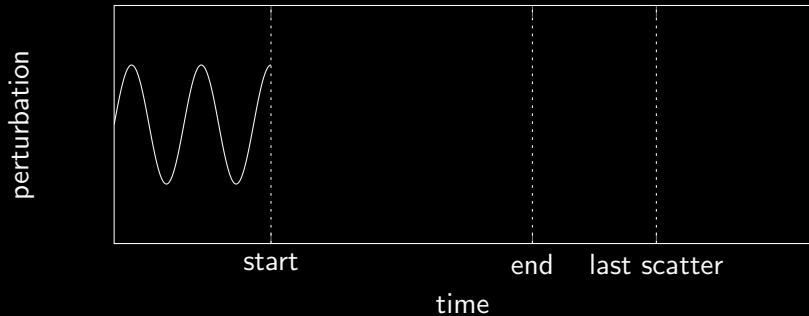
# The Planck Power Spectrum

A perfect fit



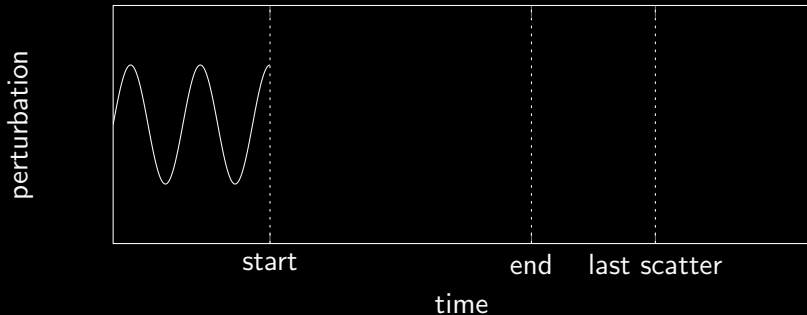
# Explaining the wiggles

Inflation at its best



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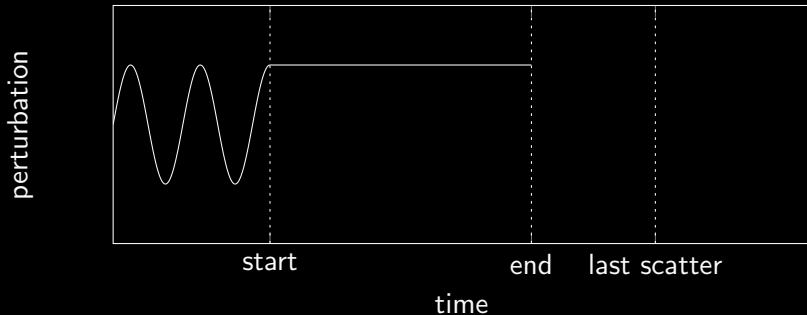
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- Inflation acts to stretch perturbations

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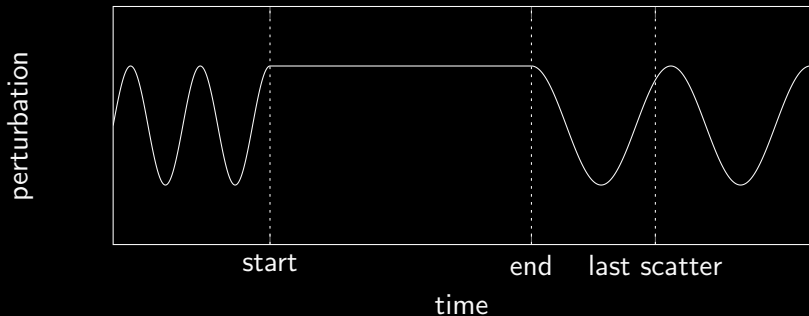
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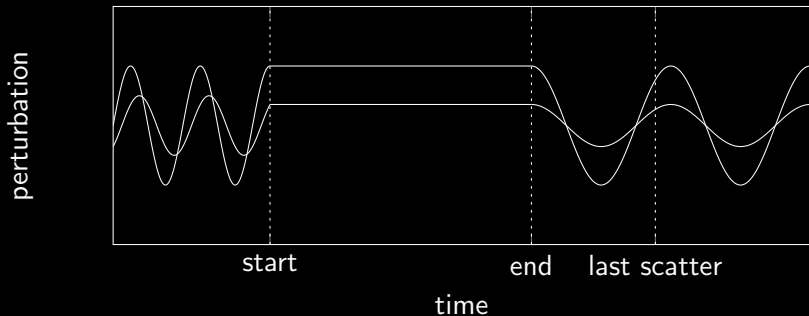
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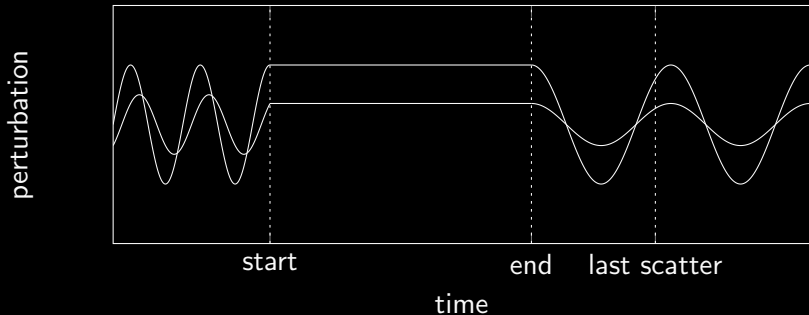
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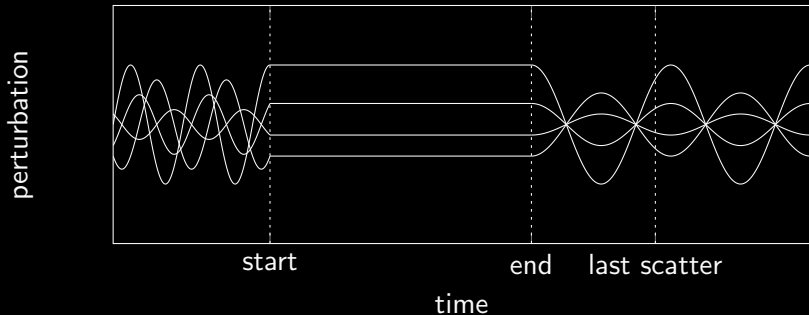


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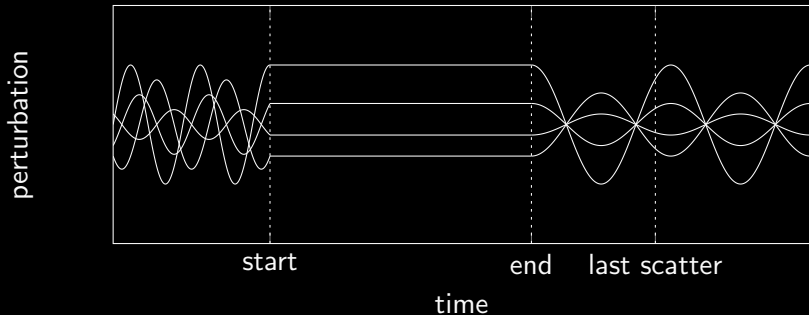
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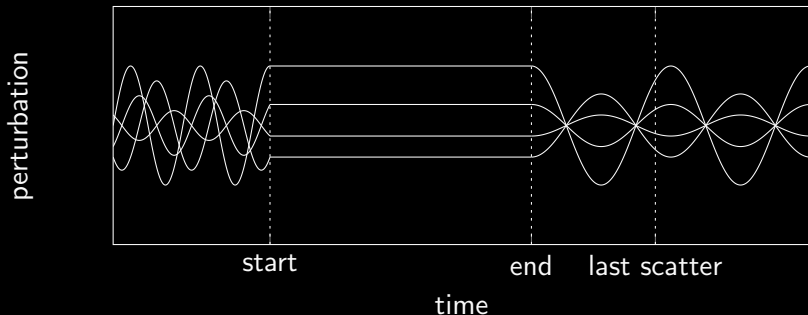
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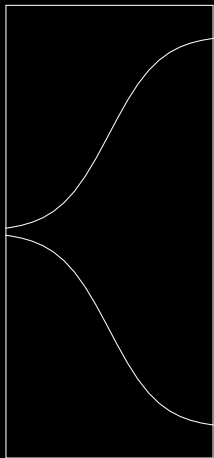
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- ▶ Last scattering surface records fixed time after inflation
- ▶ Different modes at surface will have different amplitudes

# Theory of Inflation

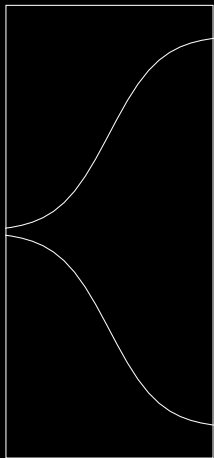
An accelerated phase



► What could generate this?

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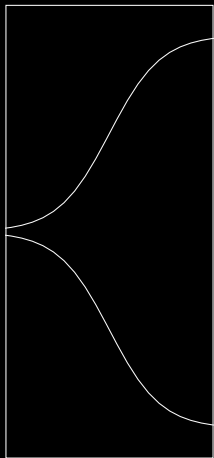
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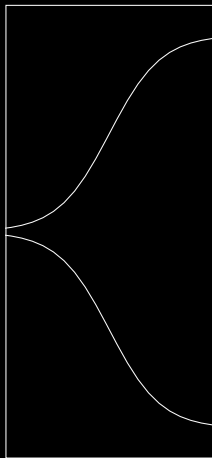
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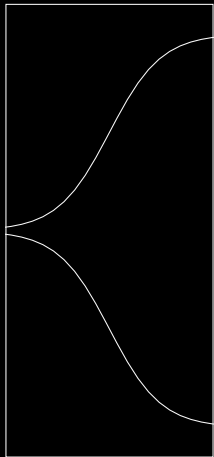
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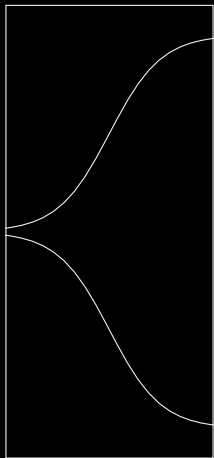


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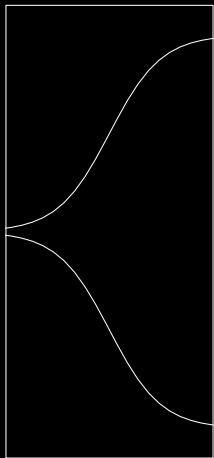
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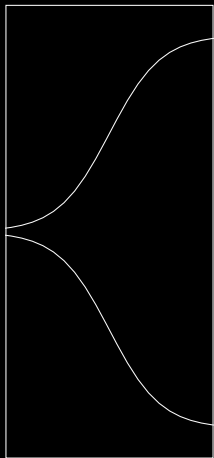
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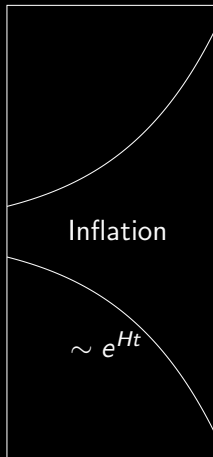
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  - ▶ Simple.
  - ▶ Cosmological Principle.
  - ▶ Effective field theory.
- ▶ The issue: We have very little idea precisely what it looks like.

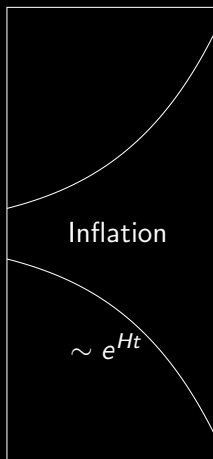
# Kinetic Dominance in the early universe

Unexpected simplicity



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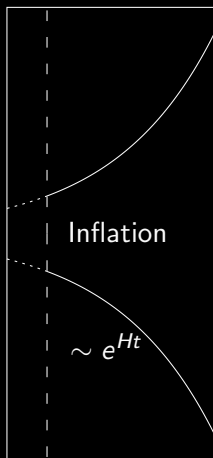
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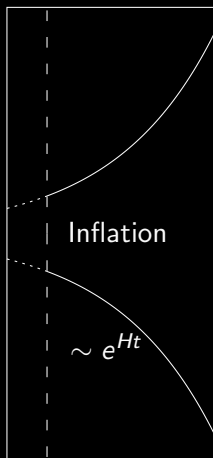
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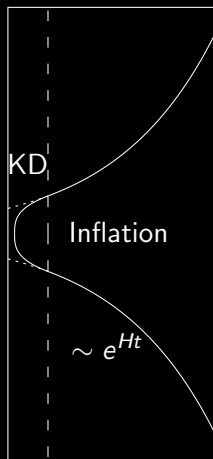
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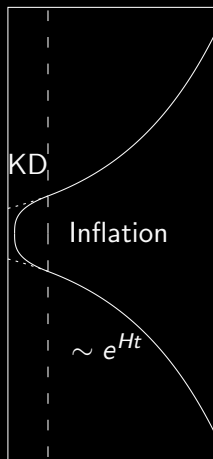


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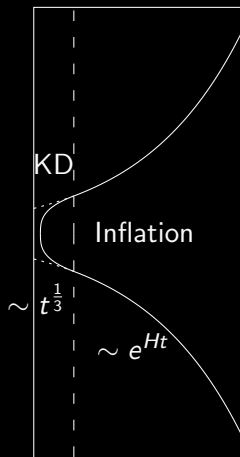
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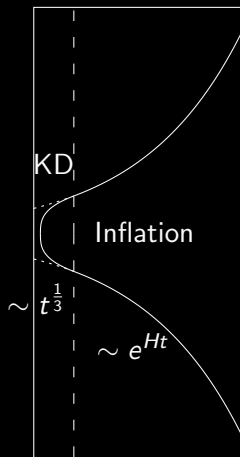
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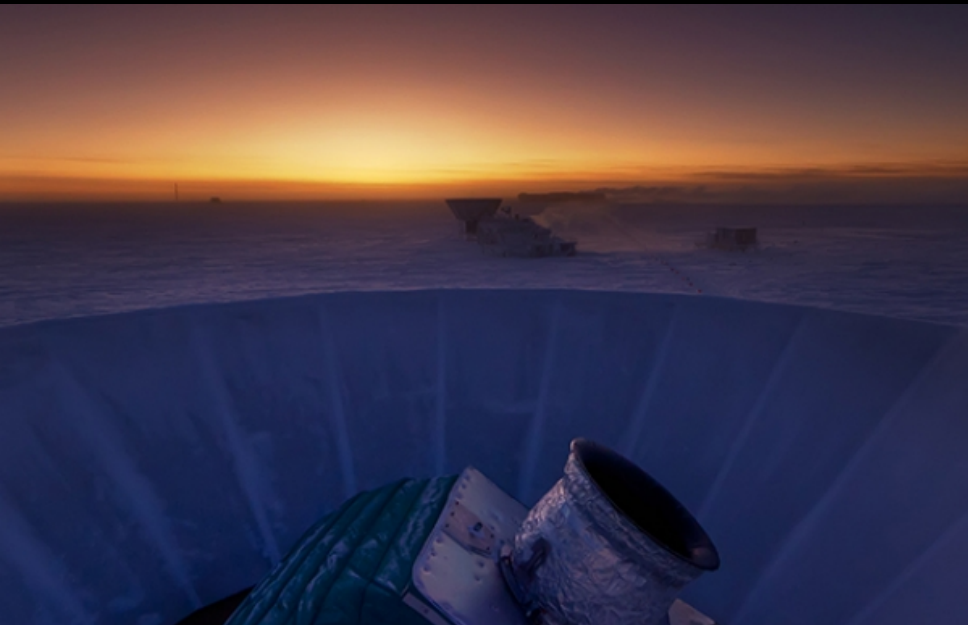
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- ▶ We have proved at early times, it doesn't matter what the fine details of the inflaton are.
- ▶ With this in hand, we are able to build extreme-early universe theory on a firm foundation.
- ▶ We are able to investigate the universe in its early, generic stage.

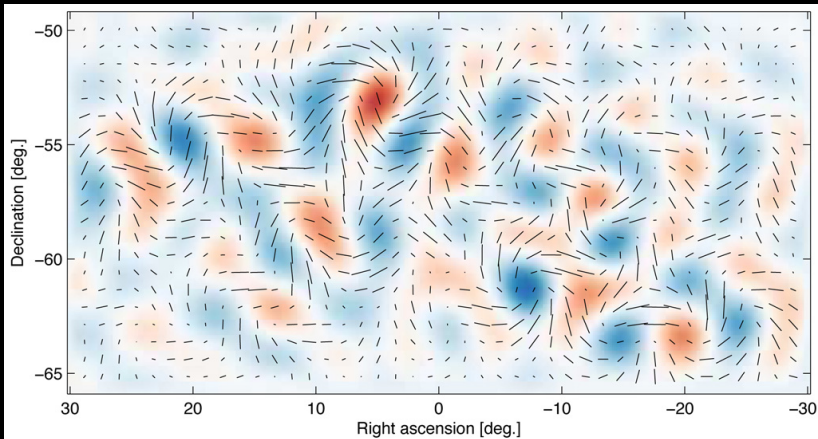
# BICEP 2

Background Imaging of Cosmic Extragalactic Polarization



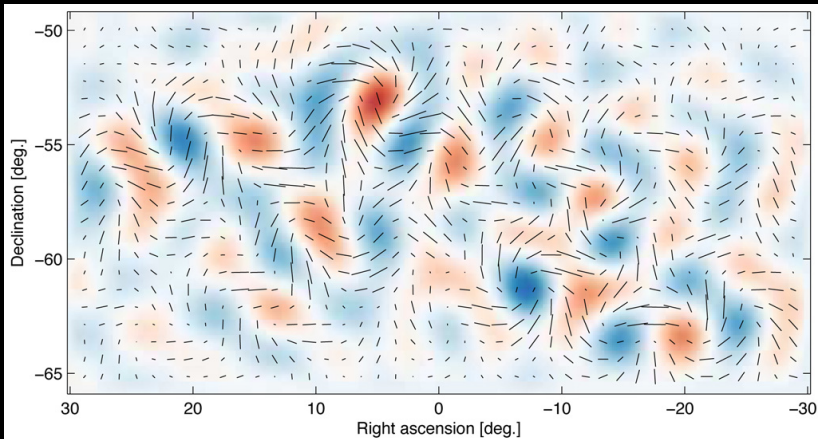
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Looking at Polarisation



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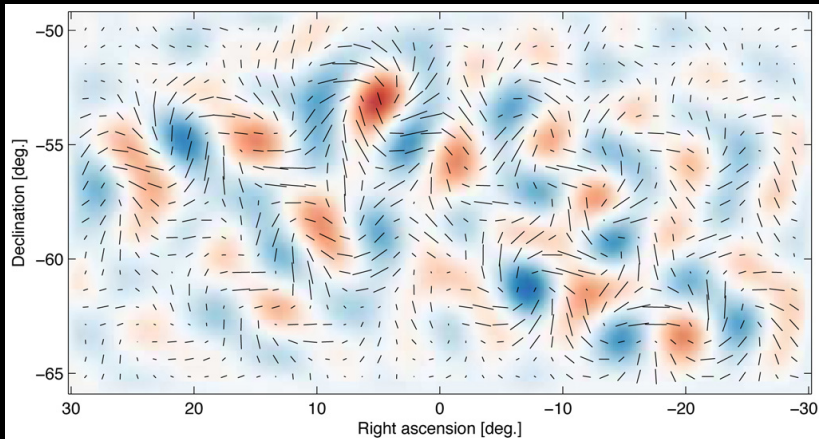
## Looking at Polarisation



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# BICEP 2

## Looking at Polarisation



- ▶ Inflation must produce gravitational waves
- ▶ BICEP have detected these primordial fluctuations.

Thank you!

Questions?

